

MUHAMMAD SOHAIL

Agricultural Engineer, AGRI/006879

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CAREER SUMMARY

Dedicated and results-driven Agricultural Engineer with hands-on experience in the Irrigation and Drainage (I&D) sector. Proficient in planning, designing, and executing advanced irrigation systems for sustainable water management. Previously served as a Research Assistant in the Agricultural Remote Sensing Lab (ARSL), UAF under the Project of NCGSA. Seeking a challenging role in a development-oriented organization to leverage expertise and contribute to impactful projects.

ACADEMIC QUALIFICATION

Bachelor of Science in Agricultural Engineering

July 2023

University of Agriculture

Faisalabad, Pakistan

CGPA: 3.29/4.0

PROFESSIONAL EXPERIENCE

Research Assistant Agricultural Engineer | January 2025 – June 2025 Agricultural Remote Sensing (ARSL) Lab, UAF | Faisalabad

Full Time

- Conducted research on crop water use, evapotranspiration estimation, and irrigation scheduling using remote sensing and GIS techniques.
- Processed and analyzed satellite imagery (Landsat, MODIS, Sentinel) for agricultural monitoring and water resource assessment.
- Applied models such as SEBAL in Google Earth Engine (GEE) to evaluate crop evapotranspiration and water use efficiency.
- Collaborated with faculty and postgraduate researchers on field data collection, lysimeter experiments, and validation studies.
- Contributed to the development of data-driven strategies for sustainable agriculture under varying climatic conditions.
- Installed and maintained a weather station and Santillometer for real-time environmental and soil moisture data collection.
- Performed geo-spatial mapping and spatial analysis for field variability, irrigation planning, and water resource management.

Internee Agricultural Engineer | June – August 2022 Water Management Training Institute Punjab, | Lahore

Full Time

- Gained practical knowledge and hands-on experience in modern water management practices.
- Conducted in-depth research and technical analysis on the design of a sustainable water storage pond.
- Applied principles of hydrology, soil analysis, and site suitability to propose efficient water storage solutions.
- Collaborated with field experts and trainers to understand real-world challenges in water conservation.

FINAL YEAR PROJECT

Design and Development of Non-Contact Water Flow Meter.

2022-2023

- Developed a non-contact smart metering device for measuring water flow rates.
- Addressed the challenge of water scarcity in agriculture and improved water use efficiency.

SUMMER INTERNSHIP PROJECT

Design of a Sustainable Water Storage Pond.

June 2022-August 2022

- Gained knowledge and hands-on experience in **water management** practices.
- Conducted research and analysis on the design of a **Sustainable Water Storage Pond** as part of the internship project.

SOFTWARE SKILLS

- Good editor skills using Microsoft Office & Minitab.
- Excellent Knowledge in 2D drafting and Basic 3d Modeling by Using AutoCAD.
- Expert in CropWAT and AquaCrop for estimation of Evapotranspiration and Irrigation Planning.
- Vast knowledge of Arduino UNO and IoT Skill for Smart Drip Irrigation design.
- Basic Knowledge in Python.
- Practical Knowledge in the Utilization of Google Earth Engine through Java Script.
- Good knowledge about the SEBAL Algorithm.
- Expert in Geographic Information System GIS (ArcGIS).
- Excellent Command in English language in Verbal & Written.
- Good Leadership, Time management & Communication Skills.
- Project Management Skill Using Microsoft Visio & Trello.

CERTIFICATION & AWARDS

- Attended Three Days International Conference on **GEO-INFORMATICS FOR WATER AND AGRICULTURE RESOURCE MANAGEMENT (ICGWARN)** on April 24-26, 2024 at the Department of Irrigation and Drainage, University of Agriculture Faisalabad.
- Attended an International Workshop on “**CLIMATE CHANGE & HYDROLOGICAL MODELING**” organized by the Department of Irrigation and Drainage, University of Agriculture Faisalabad on April 25, 2024.
- Attended the Seven Days Training Program of **Scientific Writing** Organized by Multiomics, Pakistan in December 2024.
- Awarded a Scholarship from the **Punjab Educational Endowment Fund** (Govt. Punjab)
- Successfully completed a course of **Introduction and Programming with IoT Boards** from Pohang University of Science and Technology.
- Successfully completed a course of **Water Quality for Agriculture** from University of Agriculture Faisalabad in June 2022.
- Attend One Day Seminar on **UPSCALING IRRIGATION AND CLIMATE ADVISORY SERVICES THROUGH CITIZEN SCIENCE** on November 5, 2024, at the Department of Land & Water Conservation Engineering, PMAS Agriculture University Rawalpindi.
- Awarded a Merit Scholarship from the **University of Agriculture Faisalabad**.

LANGUAGES

- English
- Urdu

PROFESSIONAL MEMBERSHIP

- **Member**, Pakistan Engineering Council (PEC).
- **Member**, The American Society of Agricultural and Biological Engineers (ASABE)
- **Member**, The American Society of Mechanical Engineers (ASME)

REFERENCES

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